UNIVERSITY OF ENGINEERING AND TECHNOLOGY, PESHAWAR PAKISTAN

Main Campus

Web Technologies

Submitted By

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Semester:	**4rd Semester, BS Computer Science**
Section:	**E**
Submission Date:	**15/3/2026**

Submitted To

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DEPARTMENT OF COMPUTER SCIENCE

UNIVERSITY OF ENGINEERING AND TECHNOLOGY, PESHAWAR, PAKISTAN

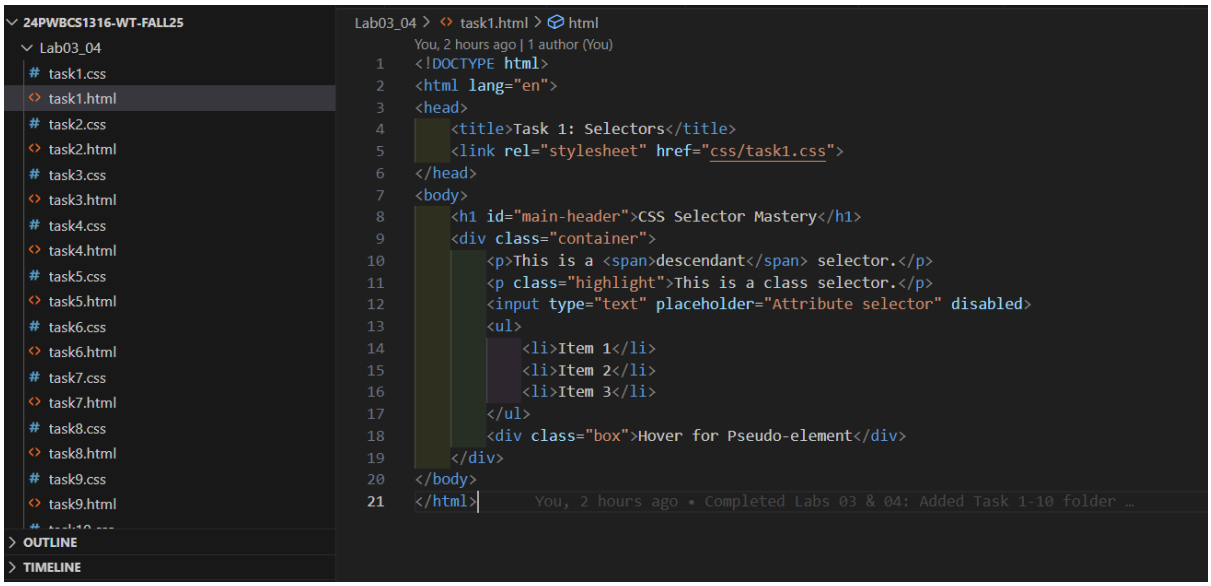
Lab 03 & 04

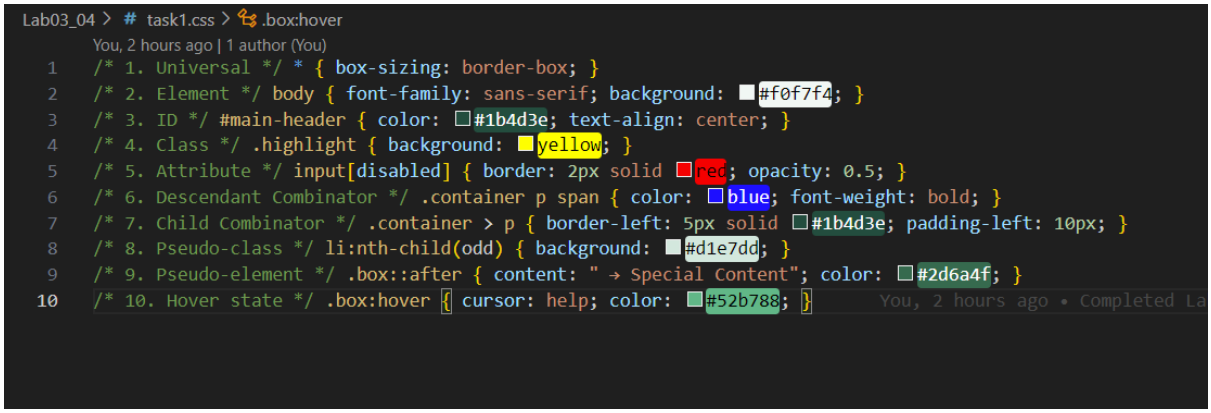
CSS Fundamentals: Styling and Visual Design

Part A: Basic Concepts & Selectors

Task 1: CSS Selectors Practice

- **Objective:** Demonstrates the ability to target HTML elements using basic, attribute, and combinator selectors.
- **Implementation:** Used ID selectors for unique headers, class selectors for repeated highlights, and child combinators for specific container styling.
- **Outcome:** Mastered CSS specificity and the hierarchy of rule application.





Task 2: Typography Showcase

- **Objective:** Focuses on modern web typography using external font libraries and scalable units.
- **Implementation:** Integrated Google Fonts (Playfair, Roboto) and applied sizing using rem and em for better accessibility.
- **Outcome:** Developed an understanding of line height, letter spacing, and text shadows for professional readability.

24PWBCS1316-WT-FALL25
Lab03_04
task1.css
task1.html
task2.css
task2.html
task3.css
task3.html
task4.css
task4.html
task5.css
task5.html
task6.css
task6.html
task7.css
task7.html
task8.css
task8.html

Lab03_04 > task2.html > html
You, 2 hours ago | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Task 2: Typography</title>
    <link rel="stylesheet" href="css/task2.css">
    <link href="https://fonts.googleapis.com/css2?family=Playfair+Display:wght@700&family=Roboto:wght
</head>
<body>
    <h1 class="google-font-1">Playfair Display (Serif)</h1>
    <p class="google-font-2">Roboto Light (Sans-Serif) - 1.2rem size.</p>
    <p class="google-font-3">Montserrat Italic - 18px size.</p>
    <h2 class="text-effect">Text Shadow & Uppercase</h2>
</body>
</html>
```

You, 2 hours ago • Completed Labs 03 & 04: Added Task 1-10 folder …

Lab03_04 > task2.css > .text-effect
You, 2 hours ago | 1 author (You)

```css
.google-font-1 { font-family: 'Playfair Display', serif; font-size: 3rem; color: #1b4d3e; }
.google-font-2 { font-family: 'Roboto', sans-serif; font-size: 1.2rem; line-height: 1.8; }
.google-font-3 { font-family: 'Montserrat', sans-serif; font-size: 18px; font-style: italic; }
.text-effect {
    text-transform: uppercase;
    letter-spacing: 5px;
    text-shadow: 2px 2px 4px rgba(0,0,0,0.3);
    color: #2d6a4f;
}
```

You, 2 hours ago • Completed Labs 03 & 04: Added Task 1-10 folder …

Task 3: Background Gallery

- **Objective:** Explores the visual depth created by background layers and gradients.
- **Implementation:** Created linear and radial gradients and applied background-blend-mode to combine images with solid colors.
- **Outcome:** Learned how to create complex textures without using high-bandwidth image files.

Lab03_04 > task3.html > html

You, 2 hours ago | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <link rel="stylesheet" href="css/task3.css">
    <title>Task 3: Backgrounds</title>
</head>
<body>
    <div class="solid">Solid Background</div>
    <div class="linear">Linear Gradient</div>
    <div class="radial">Radial Gradient</div>
    <div class="multi-layer">Multiple Layers & Blend Mode</div>
</body>
</html>
```

You, 2 hours ago • Completed Labs 03 & 04: Added Task 1

Lab03_04 > task3.css > .multi-layer

You, 2 hours ago | 1 author (You)

```css
div { width: 100%; height: 150px; margin: 10px 0; color: white; display: flex; align-items: center; justify-content: center; font-weight: bol
.solid { background-color: #1b4d3e; }
.linear { background: linear-gradient(to right, #1b4d3e, #52b788); }
.radial { background: radial-gradient(circle, #52b788, #1b4d3e); }
.multi-layer {
    background: url('../images/pattern.png'), linear-gradient(rgba(27,77,62,0.8), rgba(27,77,62,0.8));
    background-blend-mode: overlay;
    background-size: cover;
}
```

You, 2 hours ago • Completed Labs 03 & 04: Added Task 1-10 folder …

Task 4: Image Styling Portfolio

- **Objective:** Applies CSS properties to transform standard images into designed components.
- **Implementation:** Used border-radius: 50% for circular profiles and filter: grayscale() for interactive hover states.
- **Outcome:** Mastered the object-fit property to maintain aspect ratios across different container sizes.

Lab03_04 > task4.html > html

You, 2 hours ago | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <link rel="stylesheet" href="css/task4.css">
    <title>Task 4: Images</title>
</head>
<body>
    <div class="gallery">
        <img src="../images/profile.jpg" alt="Circular" class="circle">
        <img src="../images/profile.jpg" alt="Grayscale" class="filter">
        <img src="../images/profile.jpg" alt="Shadow" class="shadow">
    </div>
</body>
</html>
```

You, 2 hours ago • Completed Labs 03 & 04: Added Task 1-10 folder …

Lab03_04 > # task4.css > img

You, 2 hours ago | 1 author (You)

```css
img { width: 200px; height: 200px; margin: 20px; transition: 0.5s; object-fit: cover; }
.circle { border-radius: 50%; border: 5px solid #1b4d3e; }
.filter { filter: grayscale(100%) blur(1px); }
.filter:hover { filter: grayscale(0%) blur(0px); transform: scale(1.1); }
.shadow { box-shadow: 10px 10px 20px rgba(0,0,0,0.5); border-radius: 15px; }
```

Task 5: Button Styling Collection

- **Objective:** Demonstrates UI/UX principles by styling interactive elements based on their state.
- **Implementation:** Styled hover, active, and disabled states using transitions for smooth visual feedback.
- **Outcome:** Created a reusable library of buttons with consistent padding, shadows, and color psychology.

Lab03_04 > task5.html > html

You, 2 hours ago | 1 author (You)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <link rel="stylesheet" href="css/task5.css">
    <title>Task 5: Buttons</title>
</head>
<body>
    <button class="btn btn-primary">Primary Button</button>
    <button class="btn btn-outline">Outline Button</button>
    <button class="btn btn-danger" disabled>Disabled Button</button>
</body>
</html>
```

You, 2 hours ago • Completed Labs 03 & 04: Added Task 1-10 f

Lab03_04 > # task5.css > .btn

You, 2 hours ago | 1 author (You)

```css
.btn { padding: 12px 24px; font-size: 1rem; border-radius: 50px; cursor: pointer; transition: 0.3s; margin: 10px; border: none; }
.btn-primary { background: #1b4d3e; color: white; }
.btn-primary:hover { background: #2d6a4f; box-shadow: 0 5px 15px rgba(0,0,0,0.2); }
.btn-outline { background: transparent; border: 2px solid #1b4d3e; color: #1b4d3e; }
.btn-outline:focus { background: #d1e7dd; outline: none; }
.btn-danger:disabled { background: #ccc; cursor: not-allowed; }
```

Part B: Complex Components & Layouts

Task 6: Form Styling

- **Objective:** Improves user experience by designing clean, accessible registration forms.
- **Implementation:** Applied custom styling to text inputs, radio buttons, and select menus with high-contrast :focus states.
- **Outcome:** Learned how to use box-sizing: border-box to prevent input fields from overflowing their containers.

task6.html | ~~task6.css~~ | task7.html | task7.css | task8.html | task

Lab03_04 > task6.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Task 6: Form Styling</title>
    <link rel="stylesheet" href="css/task6.css">
</head>
<body>
    <form class="styled-form">
        <h2>Register</h2>
        <input type="text" placeholder="Full Name" required>
        <input type="email" placeholder="Email Address" required>
        <select>
            <option>Select Course</option>
            <option>Web Technologies</option>
        </select>
        <button type="submit">Submit</button>
        <p class="error-msg">Please check your input.</p>
    </form>
</body>
</html>
```

Lab03_04 > task6.css > .styled-form

You, 3 hours ago | 1 author (You)

```css
.styled-form { max-width: 400px; margin: 50px auto; padding: 20px; background: white; border-radius: 10px; box-shadow: 0 10px 25px rgba(0,0,
input, select { width: 100%; padding: 12px; margin: 10px 0; border: 1px solid #ddd; border-radius: 5px; transition: 0.3s; }
input:focus { border-color: #1b4d3e; outline: none; box-shadow: 0 0 8px rgba(27,77,62,0.2); }
button { width: 100%; padding: 12px; background: #1b4d3e; color: white; border: none; border-radius: 5px; cursor: pointer; }
.error-msg { color: #d63031; font-size: 0.8rem; display: none; }
```

Task 7: Card Component Design

- **Objective:** Combines multiple CSS properties to create a modern "Product Card" layout.

- **Implementation:** Integrated images, typography, and buttons within a container using box-shadow for depth.

- **Outcome:** Mastered the use of overflow: hidden to ensure child elements respect the card's rounded corners.

Lab03_04 > task7.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Task 7: Card Design</title>
    <link rel="stylesheet" href="css/task7.css">
</head>
<body>
    <div class="card">
        <img src="../images/profile.jpg" alt="Service">
        <div class="card-content">
            <h3>Web Development</h3>
            <p>High-quality static and dynamic websites built with modern CSS techniques.</p>
            <button class="card-btn">Learn More</button>
        </div>
    </div>
</body>
</html>
```

Lab03_04 > # task7.css > .card-btn

```css
.card { width: 300px; background: white; border-radius: 15px; overflow: hidden; box-shadow: 0 4px 15px rgba(0,0,0,0.1); transition: 0.3s; margin: 50px auto; }
.card img { width: 100%; height: 200px; object-fit: cover; }
.card-content { padding: 20px; text-align: center; }
.card:hover { transform: translateY(-10px); box-shadow: 0 15px 30px rgba(0,0,0,0.2); }
.card-btn { background: #1b4d3e; color: white; border: none; padding: 10px 20px; border-radius: 5px; cursor: pointer; }
```

Task 8: Navigation Menu Styling

- **Objective:** Creates a functional, responsive navigation system for multi-page websites.
- **Implementation:** Used flexbox for alignment and implemented a CSS-only dropdown menu using the :hover pseudo-class.
- **Outcome:** Achieved a sticky header design that remains consistent across all site sections.

Lab03_04 > task8.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Task 8: Navigation</title>
    <link rel="stylesheet" href="css/task8.css">
</head>
<body>
    <nav class="navbar">
        <div class="logo">MohsinDocs</div>
        <ul class="nav-links">
            <li><a href="#" class="active">Home</a></li>
            <li class="dropdown">
                <a href="#">Projects ▾</a>
                <ul class="dropdown-content">
                    <li><a href="#">Lab 02</a></li>
                    <li><a href="#">Lab 03</a></li>
                </ul>
            </li>
            <li><a href="#">Contact</a></li>
        </ul>
    </nav>
</body>
</html>
```

Lab03_04 > # task8.css > .dropdown:hover .dropdown-content

```css
.navbar { background: #1b4d3e; padding: 15px 50px; display: flex; justify-content: space-between; align-items: center; color: white; }
.nav-links { list-style: none; display: flex; gap: 20px; }
.nav-links a { color: white; text-decoration: none; padding: 10px; transition: 0.3s; }
.nav-links a:hover, .active { background: rgba(255,255,255,0.2); border-radius: 5px; }
.dropdown { position: relative; }
.dropdown-content { display: none; position: absolute; background: white; min-width: 150px; box-shadow: 0 8px 16px rgba(0,0,0,0.2); top: 100%; }
.dropdown-content a { color: #333; display: block; }
.dropdown:hover .dropdown-content { display: block; }
```

Task 9: CSS Transform Gallery

- **Objective:** Visualizes 2D and 3D space manipulation of HTML elements.
- **Implementation:** Applied rotate, scale, skew, and translate properties triggered by mouse interactions.
- **Outcome:** Understood the transform-origin property and its effect on rotational animations.

Lab03_04 > task9.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <link rel="stylesheet" href="css/task9.css">
    <title>Task 9: Transforms</title>
</head>
<body>
    <div class="box rotate">Rotate</div>
    <div class="box scale">Scale</div>
    <div class="box skew">Skew</div>
    <div class="box translate">Translate</div>
</body>
</html>
```

Lab03_04 > task9.css > .translate:hover

```css
.box { width: 100px; height: 100px; background: #52b788; margin: 50px; display: flex; align-items: center; justify-content: center; color: white; transition:
.rotate:hover { transform: rotate(360deg); }
.scale:hover { transform: scale(1.5); }
.skew:hover { transform: skewX(20deg); }
.translate:hover { transform: translateX(50px); }
```

Task 10: Advanced Effects Showcase

- **Objective:** Implements high-end CSS features like animations and clipping masks.
- **Implementation:** Created a "Gradient Text" effect using background-clip: text and built a CSS loading spinner with @keyframes.
- **Outcome:** Mastered the use of standard and vendor-prefixed properties for cross-browser compatibility.

Lab03_04 > task10.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <link rel="stylesheet" href="css/task10.css">
    <title>Task 10: Advanced Effects</title>
</head>
<body>
    <h1 class="gradient-text">Gradient Animated Text</h1>
    <div class="loader"></div>
</body>
</html>
```

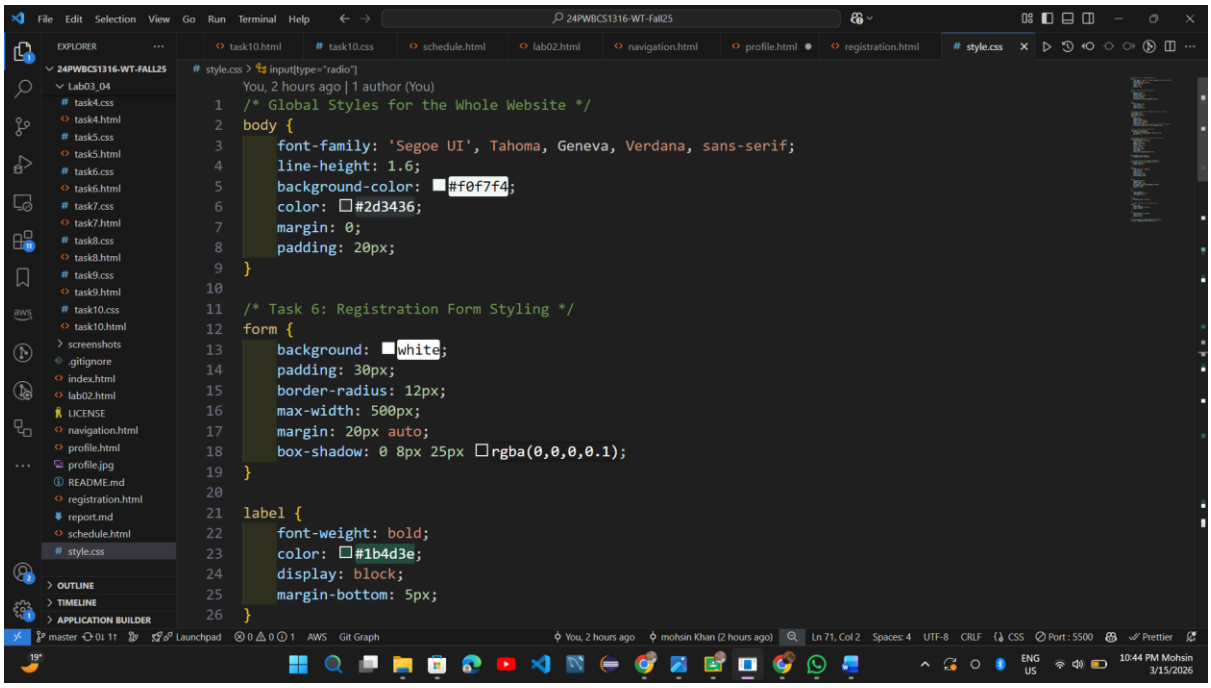
Lab03_04 > task10.css > @keyframes spin

```css
.gradient-text {
    font-size: 4rem;
    background: linear-gradient(to right, #1b4d3e, #52b788, #d1e7dd);
    background-size: 200% auto; /* Add this so the gradient can move */
    -webkit-background-clip: text;
    background-clip: text; /* Add this for standard compatibility */
    -webkit-text-fill-color: transparent;
    animation: slide 3s infinite linear;
}

@keyframes slide { from { background-position: 0%; } to { background-position: 100%; } }

.loader {
    width: 50px;
    height: 50px;
    border: 5px solid #f3f3f3;
    border-top: 5px solid #1b4d3e;
    border-radius: 50%;
    animation: spin 1s linear infinite;
}

@keyframes spin { 0% { transform: rotate(0deg); } 100% { transform: rotate(360deg); } }
```

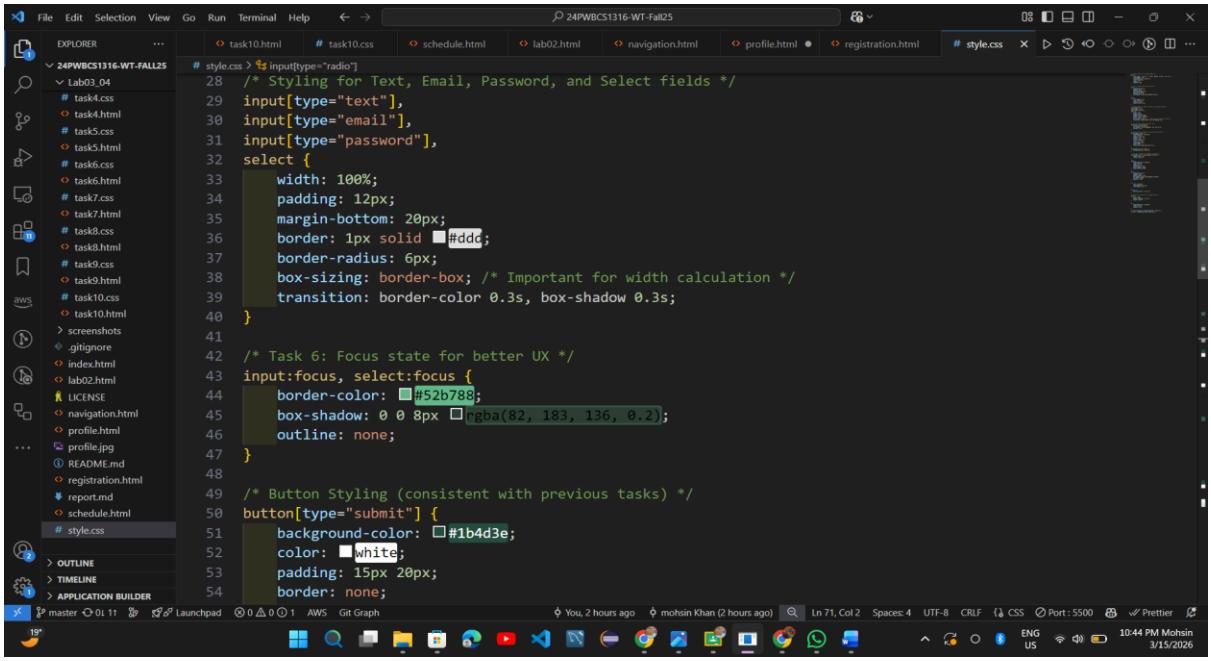
Part C: Practical Application

Task 11: Personal Portfolio Styling

- **Objective:** Refines the Lab 02 profile into a professional personal brand.
- **Implementation:** Applied a unified "Executive Luxe" color palette (Forest Green/Charcoal) using External CSS.
- **Outcome:** Ensured visual consistency across the entire website structure.



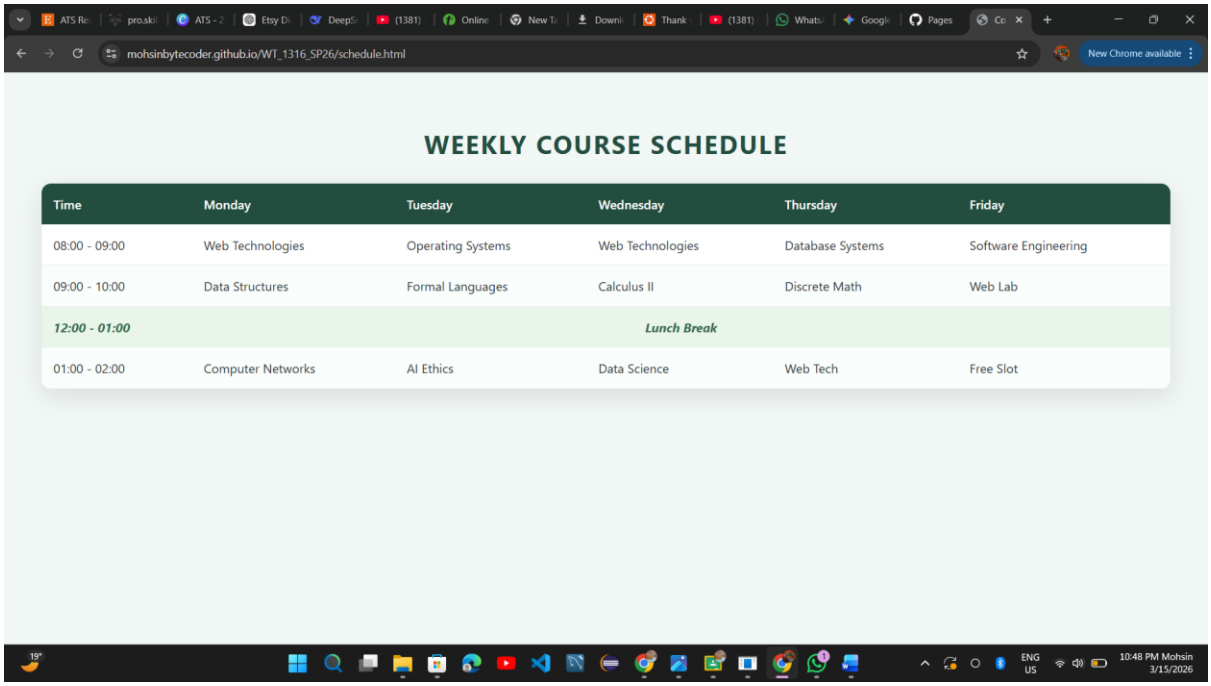
```
/* Global Styles for the Whole Website */
body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    line-height: 1.6;
    background-color: #f0f7f4;
    color: #2d3436;
    margin: 0;
    padding: 20px;
}

/* Task 6: Registration Form Styling */
form {
    background: white;
    padding: 30px;
    border-radius: 12px;
    max-width: 500px;
    margin: 20px auto;
    box-shadow: 0 8px 25px rgba(0,0,0,0.1);
}

label {
    font-weight: bold;
    color: #1b4d3e;
    display: block;
    margin-bottom: 5px;
}
```



```
/* Styling for Text, Email, Password, and Select fields */
input[type="text"],
input[type="email"],
input[type="password"],
select {
    width: 100%;
    padding: 12px;
    margin-bottom: 20px;
    border: 1px solid #ddd;
    border-radius: 6px;
    box-sizing: border-box; /* Important for width calculation */
    transition: border-color 0.3s, box-shadow 0.3s;
}

/* Task 6: Focus state for better UX */
input:focus, select:focus {
    border-color: #52b788;
    box-shadow: 0 0 8px rgba(82, 183, 136, 0.2);
    outline: none;
}

/* Button Styling (consistent with previous tasks) */
button[type="submit"] {
    background-color: #1b4d3e;
    color: white;
    padding: 15px 20px;
    border: none;
```

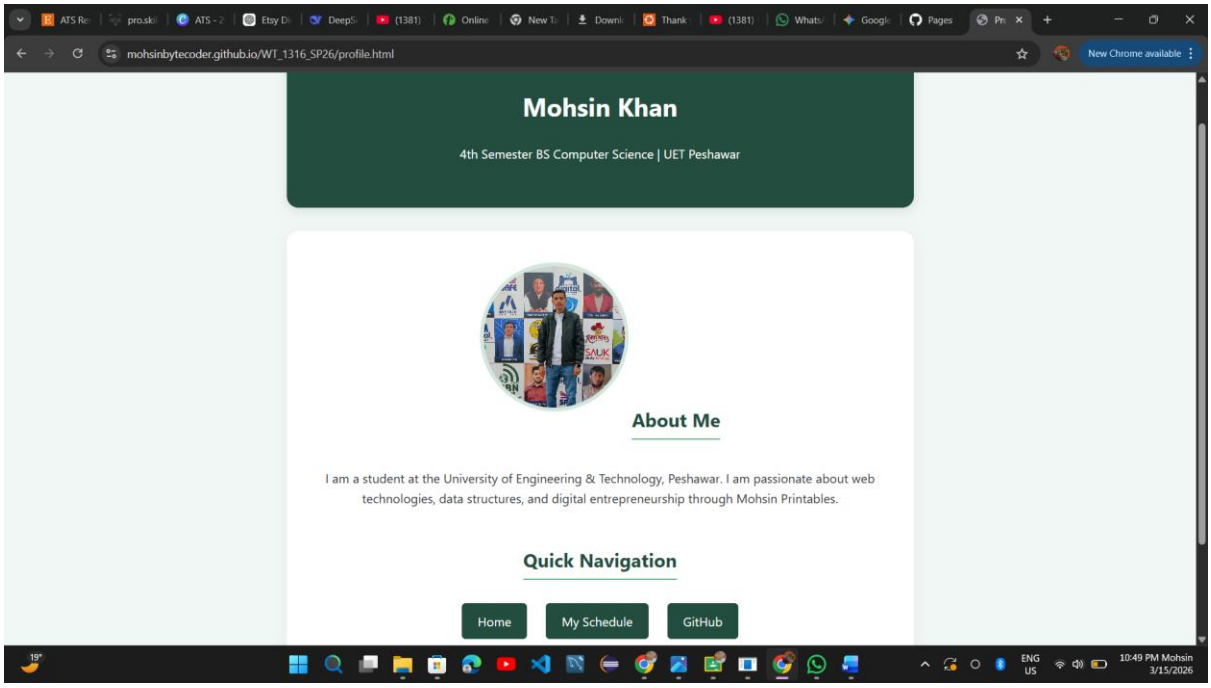
Task 12: Course Website Styling

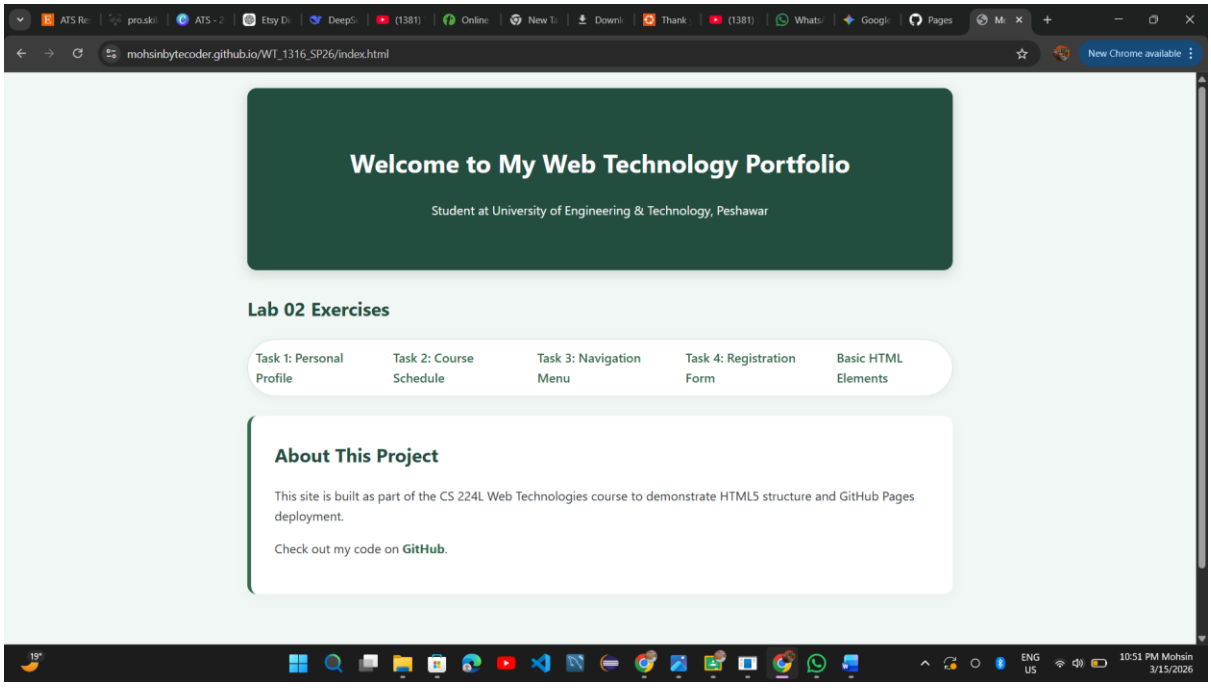
- **Objective:** Converts raw data tables into readable, interactive schedules.
- **Implementation:** Applied zebra-striping via nth-child(even) and added row-level hover highlights.
- **Outcome:** Improved data scannability and professional presentation of academic information.

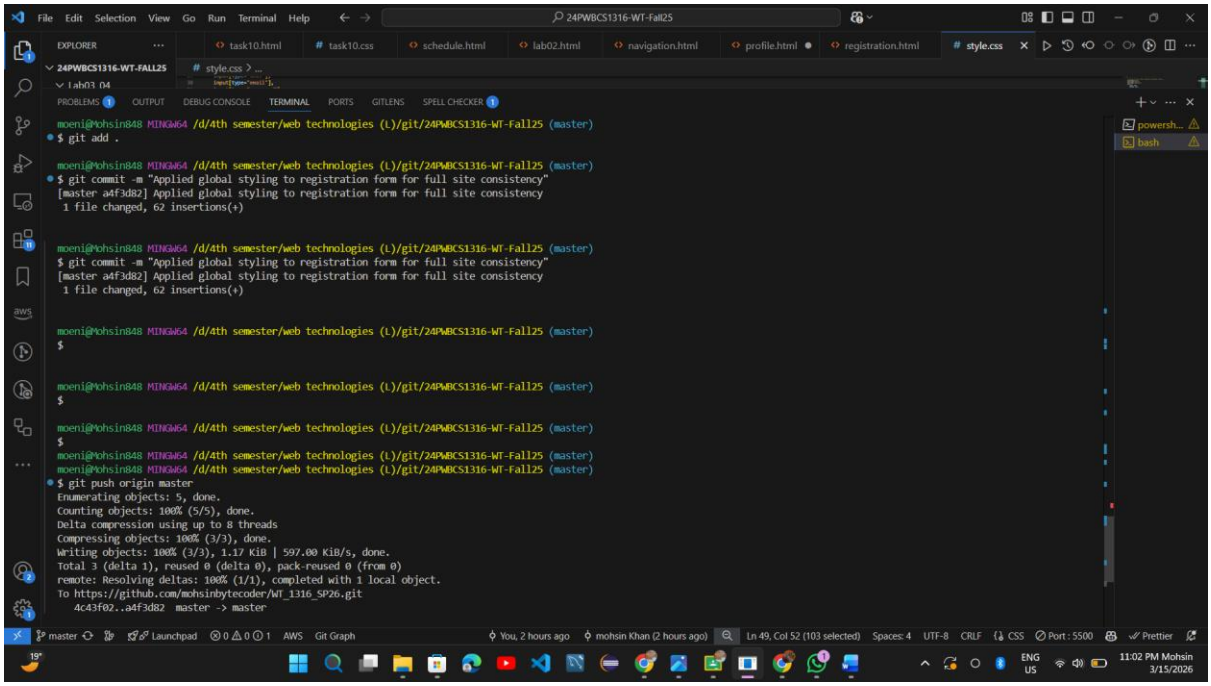


Time	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:00	Web Technologies	Operating Systems	Web Technologies	Database Systems	Software Engineering
09:00 - 10:00	Data Structures	Formal Languages	Calculus II	Discrete Math	Web Lab
12:00 - 01:00			*Lunch Break*		
01:00 - 02:00	Computer Networks	AI Ethics	Data Science	Web Tech	Free Slot

PROFILE PAGE :





Task 13: GitHub Pages Enhancement

- **Objective:** Finalizes the development lifecycle through version control and hosting.
- **Implementation:** Synchronized all project assets to the cloud using git push origin main.
- **Outcome:** Successfully deployed a multi-page, fully-styled website accessible via a public URL.